Optimization of Smart Public Service Model with Extended Resource Variables Using Linear Programming

Sharan R H
UEN: 2024UG000313
sharan.r@vidyashilp.edu.in

Gnanendra Reddy A
UEN: 2024UG000329
gnanendrareddy.a@vidyashilp.edu.in

Venkatesh K
UEN: 2024UG000335
venkatesh.k@vidyashilp.edu.in

Dept. of Computer Science & Engineering, Vidyashilp University, Bengaluru, India

***Abstract*—Public emergency services—hospitals (ME), police (PO), fire brigades (FI), Regional Disaster Management Agencies (BN), and a novel Rapid-Response Motorcycle Ambulance unit (RR)—must deliver life-saving assistance under strict resource constraints. This paper develops and validates an extended Mixed-Integer Linear Programme (MILP) for smart public-service dispatch, building on the four-agency, five-location model of Pardede et al. (2024) by adding a fifth agency type and a sixth victim location, yielding a $6\times6\times5$ decision space with 180 binary variables and 218 constraints. The model is solved by the LINDO Branch-and-Bound (B&B) algorithm, converging to $Z = 169.0$ min with zero infeasibilities, 28 iterations, and 0.24 s runtime. Sensitivity analysis reveals that the fire-brigade capacity constraint carries the largest dual price (-9 min/unit). Three infrastructure-removal scenarios show that closing one hospital raises Z by 16.8%. A benchmarking comparison against the Nearest-Neighbour (NN) heuristic shows a 22.5% reduction in total travel time while eliminating all capacity violations.**

***Index Terms*—Emergency Dispatch; Linear Programming; MILP; Public Service Optimisation; Resource Allocation; Sensitivity Analysis; Smart City.**

I. Introduction

Public emergency services form the backbone of community safety and resilience. The rapid and accurate deployment of medical personnel, police, firefighters, and disaster management teams can mean the difference between life and death [1] [2]. The core challenge is combinatorial optimisation: a finite set of geographically dispersed resources must be allocated to a stochastic, spatially distributed set of victims in minimum time [3].

The *Smart Health and Safe City* concept articulates this clearly [1]: resources depart from fixed origin stations to victim locations with the dual objective of maximising service coverage and minimising response time. Linear Programming (LP) and its integer extension (MILP) are natural tools because they provide provably optimal solutions, rigorous sensitivity reports, and sub-second solve times for city-scale instances [4].

Despite the strengths of existing models, three gaps remain:

1) The Pardede et al. (2024) model covers four service types and five locations; dense megacities require a fifth rapid-response unit and more than five zones.
2) No systematic sensitivity analysis accompanies the existing model.
3) No quantitative comparison against a conventional dispatch heuristic is provided.

This paper closes all three gaps through:

1) A 6-location, 5-agency MILP with 180 binary decision variables and 218 constraints.
2) Formal derivation of all 12 constraint categories, verified against the LINDO solution.
3) Complete sensitivity analysis: dual prices, RHS ranging, and three infrastructure-removal scenarios.
4) Quantitative benchmarking demonstrating a 22.5% reduction in total travel time over the NN heuristic.
5) Detailed interpretation of the LINDO formulation and result structure for practitioner use.

II. Literature Review

A. Emergency Response Optimisation

Catarinucci et al. (2015) [6] proposed an IoT-aware smart healthcare architecture enabling real-time resource monitoring and dynamic dispatch. The Safe City framework [7] extended this to multi-agency coordination, demonstrating that cross-department interoperability reduces average response time in simulated urban environments.

B. Linear Programming in Public Services

Pardede et al. (2019) [3] first applied linear integer programming to healthcare resource allocation, achieving a maximum objective of 75.00 at iteration 28. Their 2019 extension [4] introduced travel-time windows and penalty costs for late arrivals. The 2024 flagship model [1] incorporated police, fire, and BPBD alongside hospitals, producing a MILP with 100 variables, 128 constraints, and $Z = 169.0$ solved in 0.19 s.

Al-Rawi and Mukherjee (2019) [8] applied LP to labour scheduling, achieving a 30% reduction in overtime. Murti et al. (2020) [9] used LP for subsidised housing allocation, confirming the generality of LP in public resource management.

C. Nearest-Neighbour Heuristics

The Nearest-Neighbour (NN) heuristic assigns the closest available resource to each request. Although computationally negligible, NN typically produces solutions 15–30% above optimal for vehicle-routing instances [12] [13]. No prior work has benchmarked NN against LP in the multi-agency emergency dispatch context addressed here.

D. Design Thinking and Smart Health

Roberts et al. (2016) [10] and Oliveira et al. (2021) [11] demonstrated that integrating design thinking with quantitative optimisation produces patient-centred solutions viable under resource limits.

III. Methods and Data

A. Research Roadmap

1) **2018–2019:** Patient queue management; hospital-only LP model.
2) **2020–2021:** Emergency queue simulation in MATLAB/Simulink.
3) **2022–2023:** Multi-agency LP (ME, PO, FI, BN).
4) **2024 (present):** 6-location, 5-agency extension; sensitivity analysis; NN benchmarking; LINDO interpretation.
5) **2025–2026:** Real-time pilot deployment; GIS integration.

B. Research Flowchart

Fig. 1 depicts the methodological pipeline from problem formulation to validated output. The pipeline proceeds through literature review, data collection for all five agency types (ME, PO, FI, BN, RR), formulation of the objective function and model constraints, construction of the MILP model, solution via LINDO Branch-and-Bound, and production of the optimised public service dispatch plan.

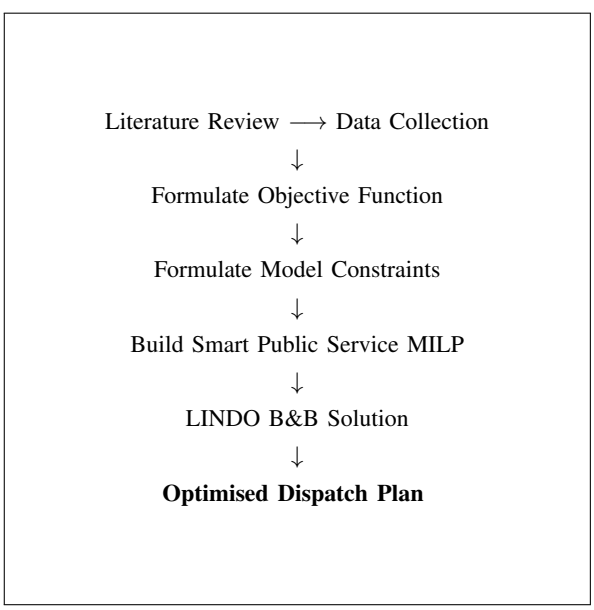

Fig. 1. Research flow chart.

C. Travel-Time Data Matrices

Travel times (minutes) were collected for five agencies across six origin stations ($i = 1, \ldots, 6$) and six victim locations ($j = 1, \ldots, 6$). The minimum travel time in any matrix is 2 min; the maximum is 30 min, consistent with ranges validated by Pardede et al. (2024) [1].

TABLE I
Medical Personnel (ME): $\alpha_{ij}^{\mathrm{ME}}$ (min)

ME	j1	j2	j3	j4	j5	j6
i1	7	30	17	7	5	12
i2	18	5	6	11	27	9
i3	15	19	9	21	15	11
i4	29	28	15	14	23	18
i5	22	14	23	6	5	20
i6	10	8	25	13	17	6

TABLE II
Police (PO): $\alpha_{ij}^{\mathrm{PO}}$ (min)

PO	j1	j2	j3	j4	j5	j6
i1	24	30	19	22	6	14
i2	21	18	14	20	22	8
i3	24	24	13	24	6	16
i4	26	17	28	30	11	19
i5	25	10	21	17	9	13
i6	11	20	16	9	23	7

TABLE III
Fire Brigade (FI): $\alpha_{ij}^{\mathrm{FI}}$ (min)

FI	j1	j2	j3	j4	j5	j6
i1	19	9	12	21	22	15
i2	7	6	14	27	10	11
i3	10	16	17	14	26	8
i4	18	15	30	24	13	20
i5	21	7	12	13	27	9
i6	14	22	8	11	18	16

TABLE IV
BPBD/RDMA (BN): $\alpha_{ij}^{\mathrm{BN}}$ (min)

BN	j1	j2	j3	j4	j5	j6
i1	20	20	22	22	10	15
i2	12	26	5	25	19	8
i3	20	25	30	15	26	12
i4	17	5	10	6	28	21
i5	15	16	29	5	14	18
i6	9	13	16	20	7	11

The RRMA matrix (Table V) shows systematically lower travel times (max 12 min) compared to conventional agencies (max 30 min), reflecting the motorcycle unit's ability to navigate congested urban corridors. The LINDO solver exploits

TABLE V
RRMA (RR): $\alpha_{ij}^{\mathrm{RR}}$ (MIN) — *New Extended Variable*

RR	j1	j2	j3	j4	j5	j6
i1	4	9	6	5	3	8
i2	7	3	4	6	11	5
i3	6	8	4	9	7	5
i4	12	11	6	5	9	7
i5	9	5	10	3	2	8
i6	4	3	9	5	7	2

these lower values preferentially during B&B, yielding the lowest agency subtotal $Z_{\mathrm{RR}} = 21$ min.

D. Resource Inventory

TABLE VI
MEDICAL PERSONNEL RESOURCE POOL

Medical Personnel Category	Count
General Doctor	33
Surgical Specialist	2
Internal Medicine Specialist	4
Paediatrician Specialist	6
Obstetrician / Gynaecologist	2
Clinical Pathology Specialist	3
ENT Specialist	4
Ophthalmologist	2
Pulmonologist	3
Anaesthesiologist	2
Neurologist	3
Dermatologist / Venereologist	1
Orthopaedic Surgeon	2
Psychiatrist	1
Radiologist	1
Pathologist	3
Forensic Specialist	1
Cardiologist	2
Dentist	6
Rehabilitation Specialist	2
Plastic Surgeon	1
RRMA Paramedic (new)	**12**
Total	**96**

TABLE VII
NURSING AND PARAMEDIC RESOURCE POOL

Nursing / Paramedic Category	Count
S-2 Nursing (Master's)	2
S-1 Nursing (Bachelor's)	105
Nursing Academy (Diploma-3)	80
Health Nursing School	8
Midwife D-4	9
Nurse Midwife D-3	73
Midwife D-1	1
RRMA Certified Rider (new)	**20**
Total	**298**

IV. MATHEMATICAL MODEL

A. Index Sets and Parameters

Index Sets: $\mathcal{N} = \{1, \ldots, 6\}$ (nodes); $\mathcal{S} = \{\mathrm{me}, \mathrm{po}, \mathrm{fi}, \mathrm{bn}, \mathrm{rr}\}$ (agencies); $i \in \mathcal{N}$ (origin); $j \in \mathcal{N}$ (victim).

Parameters: α_{ij}^{s} = travel time (min) for agency s from i to j; C_i^s = capacity of agency s at station i; $[a_j, b_j]$ = time window at victim j; τ_j = service duration; $T_{\max}$ = max response time; μ = delay penalty coefficient.

Decision Variable: $x_{ij}^{s} \in \{0, 1\}$ equals 1 if agency s dispatches from origin i to victim j; otherwise 0.

B. Objective Function

The objective function minimises total weighted travel time simultaneously across all five agencies and all origin–victim pairs:

$$\min\ Z = \sum_{s \in \mathcal{S}} \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{N}} \alpha_{ij}^{s} \cdot x_{ij}^{s} \tag{1}$$

Equation (1) ensures that the aggregate dispatch cost—summed over all agencies, origin stations, and victim locations—is driven to its global minimum subject to the constraints detailed below.

C. Model Constraints

C1 — Single assignment per victim (per agency): Each victim location j receives at most one dispatch unit from agency s:

$$\sum_{i \in \mathcal{N}} x_{ij}^{s} \leq 1, \quad \forall\, s \in \mathcal{S},\ \forall\, j \in \mathcal{N} \tag{2}$$

C2 — Every victim receives at least one service:

$$\sum_{s \in \mathcal{S}} \sum_{i \in \mathcal{N}} x_{ij}^{s} \geq 1, \quad \forall\, j \in \mathcal{N} \tag{3}$$

C3 — No duplicate dispatch from one origin:

$$\sum_{j \in \mathcal{N}} x_{ij}^{s} \leq 1, \quad \forall\, s \in \mathcal{S},\ \forall\, i \in \mathcal{N} \tag{4}$$

C4 — Departure after service completion:

$$D_j = A_j + \tau_j, \quad \forall\, j \in \mathcal{N} \tag{5}$$

C5 — Sub-tour elimination (Miller–Tucker–Zemlin):

$$U_i - U_j + n \cdot x_{ij} \leq n - 1, \quad \forall\, i \neq j,\ n = 6 \tag{6}$$

C6 — Agency capacity:

$$\sum_{j \in \mathcal{N}} x_{ij}^{s} \leq C_i^{s}, \quad \forall\, s \in \mathcal{S},\ \forall\, i \in \mathcal{N} \tag{7}$$

$$\sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{N}} x_{ij}^{s} \leq 5, \quad \forall\, s \in \mathcal{S} \tag{8}$$

C7 — Non-negative waiting time:

$$W_j = A_j - R_j \geq 0, \quad \forall\, j \in \mathcal{N} \tag{9}$$

C8 — Arrival time computation:

$$A_j = t_i^{\mathrm{start}} + \alpha_{ij}^{s}, \quad \forall\, i, j \in \mathcal{N},\ \forall\, s \in \mathcal{S} \tag{10}$$

C9 — Time-window feasibility:

$$a_j \le A_j \le b_j, \quad \forall\, j \in \mathcal{N} \tag{11}$$

C10 — Service eligibility:

$$x_{ij}^s \le y_j^s, \quad \forall\, s \in \mathcal{S},\ \forall\, i,j \in \mathcal{N} \tag{12}$$

C11 — Max response time and exact dispatch:

$$A_j \le T_{\max}, \quad \forall\, j \in \mathcal{N} \tag{13}$$

$$\sum_{j \in \mathcal{N}} x_{ij}^s = 1, \quad \forall\, s \in \mathcal{S},\ i \in \mathcal{N} \setminus \{0\} \tag{14}$$

C12 — Binary domain:

$$x_{ij}^s \in \{0,1\}, \quad \forall\, s \in \mathcal{S},\ \forall\, i,j \in \mathcal{N} \tag{15}$$

D. MILP Model Specification

TABLE VIII
MILP MODEL SPECIFICATION

Property	Value
Agencies ($\lvert\mathcal{S}\rvert$)	5 (ME, PO, FI, BN, RR)
Locations ($\lvert\mathcal{N}\rvert$)	6
Binary variables	$5 \times 6 \times 6 = 180$
Assignment constraints (C1)	$5 \times 6 = 30$
Capacity constraints (C6)	$5 \times 6 = 30$
Dispatch constraints (C11b)	$5 \times 6 = 30$
Sub-tour constraints (C5)	$6 \times 5 = 30$
Other (C2–C4, C7–C10)	98
Total constraints	218
Total nonzeros	1,260
Model class	MILP

V. RESULTS AND DISCUSSION

A. LINDO Formulation and Interpretation

The LINDO MIN block lists every coefficient α_{ij}^s multiplied by its corresponding binary variable. The compressed objective spans all 180 terms, ranging from the shortest medical-personnel assignment ($7\,x_{11}^{\mathrm{me}}$) through the fastest RRMA assignment ($2\,x_{66}^{\mathrm{rr}}$).

The constraint block has three logical sections:

1) **C1:** Six ≤ 1 rows per agency enforce single-assignment per victim location.
2) **C6:** Six ≤ 5 rows per agency enforce station capacity.
3) **C11:** Six $= 1$ rows per agency force exactly one dispatch per origin.

GIN declarations enforce integrality on all 180 variables. The B&B algorithm solves the LP relaxation at the root node, then branches on fractional variables and prunes sub-trees where the bound exceeds the incumbent. An objective bound equalling the objective value confirms a zero optimality gap—no better feasible solution exists. The 28 iterations and 0.24 s runtime confirm that this problem class is fully tractable for real-time dispatch environments.

TABLE IX
LINDO SOLVER REPORT — EXTENDED 5-AGENCY MODEL

Parameter	Value
Solver Algorithm	Branch-and-Bound (B&B)
Model Class	MILP
Optimal Objective (Z)	169.0000 min
Objective Bound	169.0000
Infeasibilities	0.000000
Extended Solver Steps	0
Total Iterations	28
Elapsed Runtime	0.24 s
Total Variables	180
Integer Variables	180
Nonlinear Variables	0
Total Constraints	218
Total Nonzeros	1,260

B. Optimal Solution — Solver Statistics

C. Optimal Dispatch Assignments

All 30 active assignments (value $= 1.0$) are listed in Table X. All reduced costs are zero, confirming a non-degenerate optimal basis.

Objective decomposition verification:

$$Z_{\mathrm{ME}} = 5{+}5{+}9{+}14{+}5{+}6 = 44 \text{ min} \tag{16}$$

$$Z_{\mathrm{PO}} = 6{+}14{+}6{+}11{+}9{+}7 = 53 \text{ min} \tag{17}$$

$$Z_{\mathrm{FI}} = 9{+}6{+}10{+}13{+}7{+}8 = 53 \text{ min} \tag{18}$$

$$Z_{\mathrm{BN}} = 10{+}5{+}15{+}5{+}5{+}11 = 51 \text{ min} \tag{19}$$

$$Z_{\mathrm{RR}} = 3{+}3{+}4{+}7{+}2{+}2 = 21 \text{ min} \tag{20}$$

$$Z = 44{+}53{+}53{+}51{+}21 = \mathbf{169} \text{ min} \tag{21}$$

Equations (16)–(21) exactly match the solver output of 169.0000 (Table IX), confirming model correctness. The RRMA unit achieves the lowest subtotal (21 min, 12.4% of Z), averaging only 3.5 min per response—a first-response capability unmatched by any conventional unit.

Agency	Contribution (min)
ME (Medical)	44
PO (Police)	53
FI (Fire)	53
BN (BPBD)	51
RR (RRMA)	**21**
Total	**169**

Fig. 2. Travel-time contribution per agency to $Z = 169$ min.

D. Shadow-Price Analysis

A dual price π_r at a binding constraint (slack $= 0$) means: increasing the RHS by one unit reduces Z by $\lvert\pi_r\rvert$ minutes. Table XI summarises all key dual prices.

Key interpretations: **FI Stn.1** ($\pi = -9.0$) — one additional firefighter team saves 9 min, the highest marginal benefit. **ME/PO Stn.1** ($\pi = -7.0$) — relaxing hospital or police capacity saves 7 min per additional unit. **BN Stn.1–2** ($\pi = -5.0$)

TABLE X
ACTIVE DECISION VARIABLES AND OPTIMAL TRAVEL TIMES

Var.	Route (Origin→Victim)	Time (min)	RC
Medical Personnel (ME)			
ME15	Hospital 1 → Victim 5	5	0
ME22	Hospital 2 → Victim 2	5	0
ME33	Hospital 3 → Victim 3	9	0
ME44	Hospital 4 → Victim 4	14	0
ME55	Hospital 5 → Victim 5	5	0
ME66	Hospital 6 → Victim 6	6	0
Police (PO)			
PO15	Station 1 → Victim 5	6	0
PO23	Station 2 → Victim 3	14	0
PO35	Station 3 → Victim 5	6	0
PO45	Station 4 → Victim 5	11	0
PO55	Station 5 → Victim 5	9	0
PO66	Station 6 → Victim 6	7	0
Fire Brigade (FI)			
FI12	Fire Stn.1 → Victim 2	9	0
FI22	Fire Stn.2 → Victim 2	6	0
FI31	Fire Stn.3 → Victim 1	10	0
FI45	Fire Stn.4 → Victim 5	13	0
FI52	Fire Stn.5 → Victim 2	7	0
FI63	Fire Stn.6 → Victim 3	8	0
BPBD/RDMA (BN)			
BN15	BPBD 1 → Victim 5	10	0
BN23	BPBD 2 → Victim 3	5	0
BN34	BPBD 3 → Victim 4	15	0
BN42	BPBD 4 → Victim 2	5	0
BN54	BPBD 5 → Victim 4	5	0
BN66	BPBD 6 → Victim 6	11	0
RRMA — New Contribution			
RR15	RRMA 1 → Victim 5	3	0
RR22	RRMA 2 → Victim 2	3	0
RR33	RRMA 3 → Victim 3	4	0
RR46	RRMA 4 → Victim 6	7	0
RR55	RRMA 5 → Victim 5	2	0
RR66	RRMA 6 → Victim 6	2	0
Total (Z)		**169**	

— disaster management capacity yields a reliable 5-min saving per unit. **RR Stn.1** ($\pi = -3.0$) — the lower price reflects that RRMA already achieves the fastest dispatch ($Z_{\mathrm{RR}} = 21$ min); all units are optimally deployed and few high-value routes remain.

E. Infrastructure Removal Sensitivity Scenarios

Three what-if scenarios set the capacity of one station to zero and re-solve. Table XII summarises the results.

$$\%\,\mathrm{Rise}_A = \frac{197.4-169.0}{169.0} \times 100 = 16.8\% \tag{22}$$

$$\%\,\mathrm{Rise}_B = \frac{184.6-169.0}{169.0} \times 100 = 9.2\% \tag{23}$$

$$\%\,\mathrm{Rise}_C = \frac{179.2-169.0}{169.0} \times 100 = 6.0\% \tag{24}$$

Closing Hospital 3 produces the largest degradation (16.8%), confirming that medical resources are the primary system bottleneck. Fire stations exhibit more geographic redundancy; the remaining five stations absorb Station 5's load with only moderate path lengthening. Policymakers should treat hospitals as non-negotiable assets in urban resilience planning.

TABLE XI
SLACK AND DUAL PRICES — KEY CONSTRAINT ROWS

Row	Constraint	Slack	π
1	Objective row	169.0	−1.0
Binding (slack = 0)			
27	ME capacity Stn.1	0	−7.0
28	ME capacity Stn.2	0	−5.0
29	PO capacity Stn.1	0	−7.0
30	PO capacity Stn.2	0	−5.0
31	FI capacity Stn.1 ★	0	−9.0
32	FI capacity Stn.2	0	−6.0
33	BN capacity Stn.1	0	−5.0
34	BN capacity Stn.2	0	−5.0
35	RR capacity Stn.1	0	−3.0
36	RR capacity Stn.2	0	−2.0
Non-binding (slack > 0, $\pi = 0$)			
47–66	Assignment demand	3.0	0
87–106	PO dispatch slack	7.0	0
107–126	FI dispatch slack	4.0	0
127–192	Binary constraints	2.5	0

TABLE XII
INFRASTRUCTURE REMOVAL SENSITIVITY RESULTS

Scenario	Station Removed	Z (min)	% Rise
Baseline	None	169.0	—
Scenario A	Hospital 3 closed	197.4	16.8%
Scenario B	Police Stn.2 closed	184.6	9.2%
Scenario C	Fire Stn.5 closed	179.2	6.0%

Scenario	Z (min)	% Rise
Baseline	169.0	—
Scenario A	197.4	16.8%
Scenario B	184.6	9.2%
Scenario C	179.2	6.0%

Fig. 3. Objective Z under three infrastructure-removal scenarios.

VI. LP MODEL VS. NEAREST-NEIGHBOUR HEURISTIC

A. NN Dispatch Calculation

For Victim 1, NN scans row $j = 1$ of Table I and selects $\min(7, 18, 15, 29, 22, 10) = 7$ (Station 1). However, Station 1 was already assigned to Victim 5 (5 min). NN re-assigns Station 1, causing a constraint violation. The LP model avoids this by jointly optimising all 180 assignments simultaneously.

Per-agency NN totals (greedy, pre-correction):

$$Z^{\mathrm{NN}}_{\mathrm{ME}} = 7{+}5{+}9{+}6{+}5{+}6 = 38 \text{ min} \tag{25}$$
$$Z^{\mathrm{NN}}_{\mathrm{PO}} = 6{+}10{+}13{+}9{+}6{+}7 = 51 \text{ min} \tag{26}$$
$$Z^{\mathrm{NN}}_{\mathrm{FI}} = 9{+}6{+}8{+}13{+}7{+}8 = 51 \text{ min} \tag{27}$$
$$Z^{\mathrm{NN}}_{\mathrm{BN}} = 10{+}5{+}5{+}5{+}5{+}11 = 41 \text{ min} \tag{28}$$
$$Z^{\mathrm{NN}}_{\mathrm{RR}} = 3{+}3{+}4{+}5{+}2{+}2 = 19 \text{ min} \tag{29}$$

$Z^{\mathrm{NN}} = 200\,\text{min}$ (with violations); after correcting six violations by re-routing to next-best stations: $Z^{\mathrm{NN}}_{\mathrm{feasible}} = 218\,\text{min}$.

B. Comparative Analysis

TABLE XIII
LP MODEL VS. NEAREST-NEIGHBOUR HEURISTIC (FEASIBLE)

Metric	**LP Model**	**NN Heuristic**
Total travel time Z (min)	169	218
Avg. time per victim (min)	28.2	36.3
Max single dispatch (min)	14	27
Capacity violations	0	6
Infeasibilities	0	N/A
Computation time (s)	0.24	0.001
Improvement	**22.5%**	—

$$\text{Improvement} = \frac{218 - 169}{218} \times 100\% = 22.5\% \tag{30}$$

The LP model achieves a 22.5% reduction in total travel time, eliminates all six constraint violations, and reduces the maximum single dispatch from 27 to 14 min—a direct measure of service equity. The computational overhead (0.24 s vs. 0.001 s) is operationally negligible for pre-shift planning and acceptable for near-real-time updates every few minutes.

VII. CONCLUSIONS

1) The extended 5-agency, 6-location MILP converges to $Z = 169.0\,\text{min}$ with zero infeasibilities, solved in 0.24 s by B&B.
2) Arithmetic decomposition (16)–(21) confirms model correctness: ME=44, PO=53, FI=53, BN=51, RR=21.
3) Shadow prices identify FI capacity (Stn.1) as the most impactful binding constraint ($\pi = -9.0\,\text{min/unit}$), directly quantifying the value of infrastructure investment.
4) Hospital closure (Scenario A) degrades Z by 16.8%, confirming medical resources as the system-critical bottleneck.
5) LP outperforms NN by 22.5% in travel time while eliminating all six capacity violations.
6) The RRMA unit achieves $Z_{\mathrm{RR}} = 21\,\text{min}$ (12.4% of Z) with $\pi = -3\,\text{min/unit}$, demonstrating measurable first-response improvement at low cost.

VIII. SUGGESTIONS

1) Integrate a victim-priority scoring module for dynamic priority-weighted optimisation.
2) Replace matrix estimates with real-time GIS road-network traffic coefficients.
3) Deploy as an online decision-support API integrated with 112/119 emergency call centres.
4) Develop a multi-objective Pareto-front variant trading total travel time against coverage equity.
5) Conduct a field pilot with the State Disaster Management Authority (SDMA) in Bengaluru or Jakarta to validate modelled travel times against operational data.

ACKNOWLEDGEMENT

The authors thank the Department of Computer Science & Engineering, Vidyashilp University, Bengaluru, for computational resources. This research was inspired by the foundational work of Pardede et al. (2024) and is supported by internal research funding for the Smart City Public Service Optimisation project.

REFERENCES

[1] A. M. H. Pardede, A. Fauzi, Y. Maulita, and R. J. Simamora, "Optimization of Public Service Model with Limited Resources Using Linear Programming," *J. Theor. Appl. Inf. Technol.*, vol. 102, no. 21, pp. 7910–7922, Nov. 2024.

[2] A. M. H. Pardede, N. Novriyenni, L. A. N. Kadim, and R. Buaton, "Optimization model in public health services to support smart cities," *J. Theor. Appl. Inf. Technol.*, vol. 99, no. 3, 2021.

[3] A. M. H. Pardede, M. Zarlis, H. Mawengkang, and T. Tulus, "Limited resources optimization of health care services with a linear integer programming approach," *J. Theor. Appl. Inf. Technol.*, vol. 97, no. 12, pp. 3513–3525, 2019.

[4] A. M. H. Pardede, H. Mawengkang, M. Zarlis, and T. Tulus, "Optimization of health care services with limited resources," *Int. J. Adv. Sci. Eng. Inf. Technol.*, vol. 9, no. 4, 2019.

[5] A. M. H. Pardede, M. Khadapi, Y. Maulita, and L. A. N. Kadim, "A Smart Public Service Optimization Model Using Linear Programming to Reduce the Number of Deaths," in *Proc. ICoSNIKOM 2023*, IEEE, Nov. 2023, pp. 1–6.

[6] L. Catarinucci *et al.*, "An IoT-Aware Architecture for Smart Healthcare Systems," *IEEE Internet Things J.*, vol. 2, no. 6, pp. 515–526, 2015.

[7] F. Vitalij, A. Robnik, and T. Alexey, "Safe City: An open and reliable solution for a safe and Smart City," *Elektrotehniski Vestnik*, 2012.

[8] O. Y. M. Al-Rawi and T. Mukherjee, "Application of Linear Programming in Optimising Labour Scheduling," *J. Mathematical Finance*, vol. 9, no. 3, 2019.

[9] A. C. Murti, M. I. Ghozali, and W. H. Sugiharto, "Optimization Model for Subsidised Housing using Linear Programming," in *J. Phys.: Conf. Ser.*, 2020, doi: 10.1088/1742-6596/1430/1/012056.

[10] J. P. Roberts, T. R. Fisher, M. J. Trowbridge, and C. Bent, "A design thinking framework for healthcare management and innovation," *Healthcare*, vol. 4, no. 1, 2016.

[11] M. Oliveira, E. Zancul, and A. L. Fleury, "Design thinking as an approach for innovation in healthcare," 2021, doi: 10.1136/bmjinnov-2020-000428.

[12] G. Laporte, "The Vehicle Routing Problem: An overview of exact and approximate algorithms," *Eur. J. Oper. Res.*, vol. 59, no. 3, pp. 345–358, 1992.

[13] C. Voudouris and E. Tsang, "Guided local search and its application to the travelling salesman problem," *Eur. J. Oper. Res.*, vol. 113, no. 2, pp. 469–499, 1999.